Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Based Neuromuscular Digital Biomarker Platform for Multiple Neuromuscular Disorders with Multi-Ethnic Clinical Study			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio · Medical		Bio · Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Medical Information & Systems	Remote and Home Health care Devices
Concept of Proposed Technology	The proposed technology is an **AI-based neuromuscular digital biomarker platform** that quantifies neuromuscular function using **wearable biosensing systems and advanced machine learning models** applied to neuromuscular bio-signals collected during standardized motor tasks. The platform integrates: ● High-quality acquisition and quality control of neuromuscular bio-signals in both clinical and at-home settings			

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	• AI models that generate **interpretable digital indices** to support rapid pre-screening and evaluation of multiple neuromuscular disorders, facilitating early differential diagnosis and disorder severity evaluation. • Multi-site calibration to ensure **generalizability across diverse clinical settings and neuromuscular disorder cohorts** • Robustness across heterogeneous clinical environments and diverse neuromuscular disease cohorts • The platform designed for single-assessment and multi-disorder pre-screening to prioritize likely neuromuscular disorder categories for downstream confirmatory testing The technology is intended to serve as an **objective and scalable functional assessment tool** for clinical care, rehabilitation, and therapeutic development, including use as digital endpoints in clinical trials and remote patient monitoring.
Rationale for Joint R&D Project	Neuromuscular disorders require objective, scalable, and cost-effective measures of disorder severity and progression for both clinical management and therapeutic development. Existing functional assessments and in-clinic tests are often subjective, time-intensive, and difficult to deploy frequently, limiting their utility for longitudinal monitoring and decentralized studies. Wearable, AI-derived digital biomarkers offer a promising alternative; however, **clinical trust, adoption, and regulatory readiness** depend on two critical factors: 1. **Clinical validation across diverse populations**, and 2. **Biological plausibility and mechanistic interpretability** of derived digital features. International joint R&D is essential to address both requirements. Collaboration enables access to **multi-ethnic patient populations and clinical research infrastructure** for evaluating generalizability and fairness. This joint framework strengthens scientific credibility, supports future regulatory planning, and aligns with the objectives of the TIPA International Joint R&D Program.
Key Words	1. Neuromuscular digital biomarker 2. Pre-screening for multiple neuromuscular disorders 3. AI/ML for biosignals 4. Neuromuscular disorders 5. Heterogeneous clinical settings

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (5)	**End of the Project**	Level (7)
R&D Objectives	**Overall Objective** To secure **U.S.-based multiple neuromuscular disorders pilot clinical evidence** demonstrating the analytical validity, clinical validity, and generalizability of an AI-based neuromuscular digital biomarker platform, and to advance the technology to **TRL 7** readiness. **Specific Objectives**			

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	• Define standardized neuromuscular bio-signal acquisition protocols and a task library for assessment • Develop and validate AI models for multi-disease classification and severity/progression evaluation • Quantify and mitigate performance gaps across measurement environment, disorder subtypes, and major race/ethnicity through calibration and robust training
R&D Target Specifications	Quantitative targets include: • **Analytical Validity**: Test–retest reliability with intraclass correlation coefficient (ICC) ≥ 0.85 for core indices • **Clinical Validity**: Area under the curve (AUC) ≥ 0.80 for disease versus control classification in target indications Correlation coefficient r ≥ 0.70 with reference clinical endpoints • **Generalizability and Fairness**: Performance gap across major race/ethnicity strata ≤ 5% (absolute AUC) after calibration • **Feasibility**: ≥ 90% adherence to prescribed assessment protocols Missing data rate ≤ 10% in pilot studies • **Data Governance**: Implementation of de-identification and secure cloud pipelines consistent with IRB requirements • **Efficiency**: Reduction of total assessment time by ≥ 50% compared to conventional diagnostic workflows
R&D Plan	The R&D program is structured over **36 months** in three sequential phases. **Year 1 (Months 0–12): Integration and Readiness** • Integration of wearable bio-signal acquisition systems and secure cloud pipelines • Design of standardized assessment protocols and clinical workflows • Establishment of a physiologically interpretable digital biomarker with neuromuscular functional phenotypes • Preparation of regulatory and ethics documentation and initiation of pilot recruitment **Year 2 (Months 13–24): Cohort Expansion and Model Refinement** • Expansion of U.S. multi-ethnic patient cohorts with longitudinal follow-up • Refinement of AI models for multi-disease classification and severity/progression evaluation • Implementation of artifact robustness and domain adaptation strategies

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	• Validation of disease-specific digital biomarkers **Year 3 (Months 25–36): Prospective Validation and Deliverables** • Freezing of AI models and pre-specified statistical analysis plans • Prospective validation of pre-screening performance in independent patient subsets • Compilation of pilot validation packages covering performance, generalizability, feasibility, and screening efficacy • Development of next-step plans for multi-site pivotal studies and SaMD regulatory pathways

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will focus on **clinical scientific leadership** and multi-ethnic cohort management including: • Recruitment of diverse patient cohorts across multiple neuromuscular disorders and healthy controls. • Execution of standardized bio-signal data acquisition and provision of clinical ground truth. • Medical validation comparing AI-based pre-screening results with standard clinical assessments. • Support for clinical study design, endpoint definition, and IRB documentation •
Expected R&R of Korean SME Partner	The Korean SME partner will lead **technology development and platform execution**, including: • Provision of wearable biosensing systems for single-assessment neuromuscular bio-signal acquisition • Development and operation of mobile applications, clinician dashboards, and secure data pipelines supporting pre-screening workflows • Development and validation of AI/ML models for neuromuscular digital biomarkers • Statistical analysis, reporting, and data governance • Support for participant engagement, remote assessments, and quality management readiness

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Scientific and Technical Outcomes** • Creation of a **validated neuromuscular digital biomarker platform** with demonstrated analytical and clinical validity and longitudinal monitoring capability • Establishment of standardized assessment protocols scalable to multi-site studies • Establishment of a large-scale multi-ethnic database covering multiple neuromuscular disorders.

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	Clinical and Regulatory Impact ● Enables multi-disease classification for symptomatic patients with uncertain etiology, informing confirmatory testing ● Improved readiness for use as digital endpoints in clinical trials ● Support for remote neuromuscular assessment workflows ● Foundational evidence for future Software as a Medical Device (SaMD) regulatory submissions **Industrial and Societal Impact** ● Advancement of objective, scalable neuromuscular disorders pre-screening tools accessible to primary care settings ● Strengthening of international collaboration between Korean SMEs and global research institutes ● Contribution to more equitable and accessible assessment of neuromuscular disorders across heterogeneous clinical settings

6. References

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